

# DEEP DOUBLE DESCENT: A REDUCED REPRODUCTION

## *Assignment 2: replication track*

Advanced Topics in Deep Learning, University of Copenhagen

### ABSTRACT

We study a reduced subset of experiments from the paper "Deep Double Descent: Where Bigger Models and More Data Hurt". Our experiments cover CIFAR-10 and CIFAR-100 width sweeps, training trajectories, and selected optimizer, augmentation, and sample-size comparisons.

### 1. INTRODUCTION

Nakkiran et al. [1] investigate double descent with respect to model size, training time, and sample size. Their public repository provides image-model definitions, plotting code, and links to results including detailed training logs.

### 2. METHODOLOGY AND EXPERIMENTAL DESIGN

#### 2.1. Reduced Training

To cover several experimental settings within our compute budget, we reduced Adam training from 4,000 to 400 epochs and CIFAR-10 SGD training from 500,000 to 50,000 updates. These are 10% of the original experiment; the CIFAR-100 CNN run instead uses 5% of the stated one-million SGD steps.

#### 2.2. Model implementation

The original study uses three architecture families: width-scaled ResNet18, a five-layer CNN, and a Fairseq Transformer. Our completed experiments train the two vision families from random initialization. We reused the authors' model definitions of ResNet18 and the five-layer CNN, rather than reimplementing the architectures. Both scale convolutional channels as  $[k, 2k, 4k, 8k]$ . ResNet18 uses two preactivation residual blocks per stage; the CNN uses four convolutional stages with batch normalization and ReLU, spatial max pooling, and a final linear classifier. For the NLP experiment we use the same Transformer implementation as the paper, namely the fairseq Transformer Fairseq v0.9.0. Its six-layer encoder and decoder use eight attention heads, feed-forward width  $4d$ , sinusoidal positions, and post-normalization. We wrote our own PyTorch training code, data loading, evaluation, resumable checkpointing, and concurrency primitives

that allow us to manage multiple training runs on the same GPU.

Our loader scales pixels to  $[0, 1]$  without channel normalization, samples one fixed corruption mask per seed, and replaces selected labels with uniformly sampled incorrect classes. This noise procedure follows Section 4 of the paper, which specifies that label noise is sampled once rather than at each epoch [1]. Subsets use a seeded permutation, implementing the paper's random subsampling reproducibly. Augmentation, when enabled, uses four-pixel padding, a random  $32 \times 32$  crop, and horizontal flipping, as specified in Appendix B.2. All augmentations of an image retain the same training label.

Scaling pixels by  $1/255$  is our preprocessing default; the paper and released model and plotting code do not establish whether the authors used channel normalization. This is a simple, reproducible choice that bounds the input values and keeps the same transformation across datasets, subset sizes, and noise settings without estimating additional channel statistics. We cannot establish that it matches the authors' preprocessing, and normalization could affect optimization and the resulting curves.

We evaluate the training subset without augmentation against its fixed training labels and the full 10,000-image test split against clean labels. Mean cross-entropy and misclassification fractions are recorded every ten epochs for Adam and every 1,250 updates for SGD, for comparison with the authors' unchanged numeric results.

#### 2.3. Training and optimizers

All reported runs use batch size 128 as the paper, cross-entropy loss, seed zero, FP32 training, and no weight decay. Adam runs use a constant learning rate  $10^{-4}$  with PyTorch's default moments  $(\beta_1, \beta_2) = (0.9, 0.999)$  and  $\epsilon = 10^{-8}$ . SGD uses no momentum and our explicitly implemented schedule

$$\eta_t = \frac{0.1}{\sqrt{1 + \lfloor t/512 \rfloor}},$$

where  $t$  is the number of completed updates.

This implements the paper's inverse-square-root learning rate schedule. Adam metrics and checkpoints are saved every ten epochs, and SGD metrics and checkpoints every 1,250 updates, yielding 40 measurements per completed run. Check-

points preserve model, optimizer, random-number state, and the data position for resumption after interruption.

## 2.4. Hidden Dimension size

The original paper does a sweep of each dimension from 1-128. In order to save compute we sweep a much smaller grid of 2, 3, 4, 6, 8, 12, 16, 24, 32, 64 for Resnet18 and 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64 for CNN. For the CNN trained with augmentation we use 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 36, 40, 48, 64, 96, 128

The reason behind the non uniform grid-selection comes down to certain figures relying on a specific width, that thus has to be added to the grid

We include a full comparison of our setup with the paper’s stated protocol and released code in Appendix A, Table 2, including source pages, appendix sections, and released filenames.

## 2.5. Runs

Table 1 summarizes the completed experiments. The seven jobs used 14.98 GPU-hours in total.

# 3. RESULTS

## 3.1. Model size, label noise, and optimization

The width sweeps and associated comparisons are organized by their experimental question rather than by compute cost. Paper Figures 1, 2, 4, and 5 concern width and noise or augmentation; Figures 6 and 7 add optimizer and dataset comparisons. All replicated figure families are presented as comparisons of ours with the authors in Appendix B. At 20% noise, our augmented CIFAR-10 CNN test error rises from 21.74% at width 16 to 28.11% at width 48, then falls to 25.88% at width 128 at step 50,000. This single-seed non-monotonicity does not establish the paper’s converged interpolation-threshold result.

## 3.2. Training time

Paper Figures 9 and 10 motivate comparisons of width-dependent training trajectories. Our ResNet histories cover 400 epochs and our CNN histories cover 50,000 steps. Figure 8 uses the ResNet notebook’s plotted epoch convention. Figure 9 covers only the width-128 CIFAR-10 CNN arm of paper Figure 10, not its two width-128 ResNet arms. Our references use the released clean full-data width sweep (1,952 positions) and the explicitly labeled 50,000-example, 20%-noise sample grid (976 positions), whereas ours have 40 measurements. Their logging cadence is not established, so we show authors’ saved indices and our recorded updates on separate logarithmic axes, without claiming time alignment or applying smoothing. The latter reference ends at 21.67%

test error; an earlier selected 20% archive ended at 37.05% and was not interchangeable with this full-data sample-grid reference. The 10%-noise published source has 1,421 positions and unresolved entry metadata; it is excluded from the plotted comparison. Both sides therefore show the common 0 and 20% noise settings on separate time axes. A smoother published curve alone therefore does not establish equivalent training or a failure of our reproduction.

## 3.3. Sample size

Paper Figures 11(a) and 12 motivate the sample-size comparisons. At width 16 and 20% label noise, our augmented CIFAR-10 CNN test errors are 40.97%, 31.19%, and 21.74% for 12,500, 25,000, and 50,000 training examples, respectively, at step 50,000. The corresponding train errors against corrupted labels are 10.71%, 25.89%, and 29.67%. At this width and horizon, more data improves test error; these measurements do not demonstrate the paper’s claim that more data can hurt in a critical regime. Figures 10 and 11 show the broader measured grids. Our 10%-noise sample curves have no matching published subset source and are excluded from these figures.

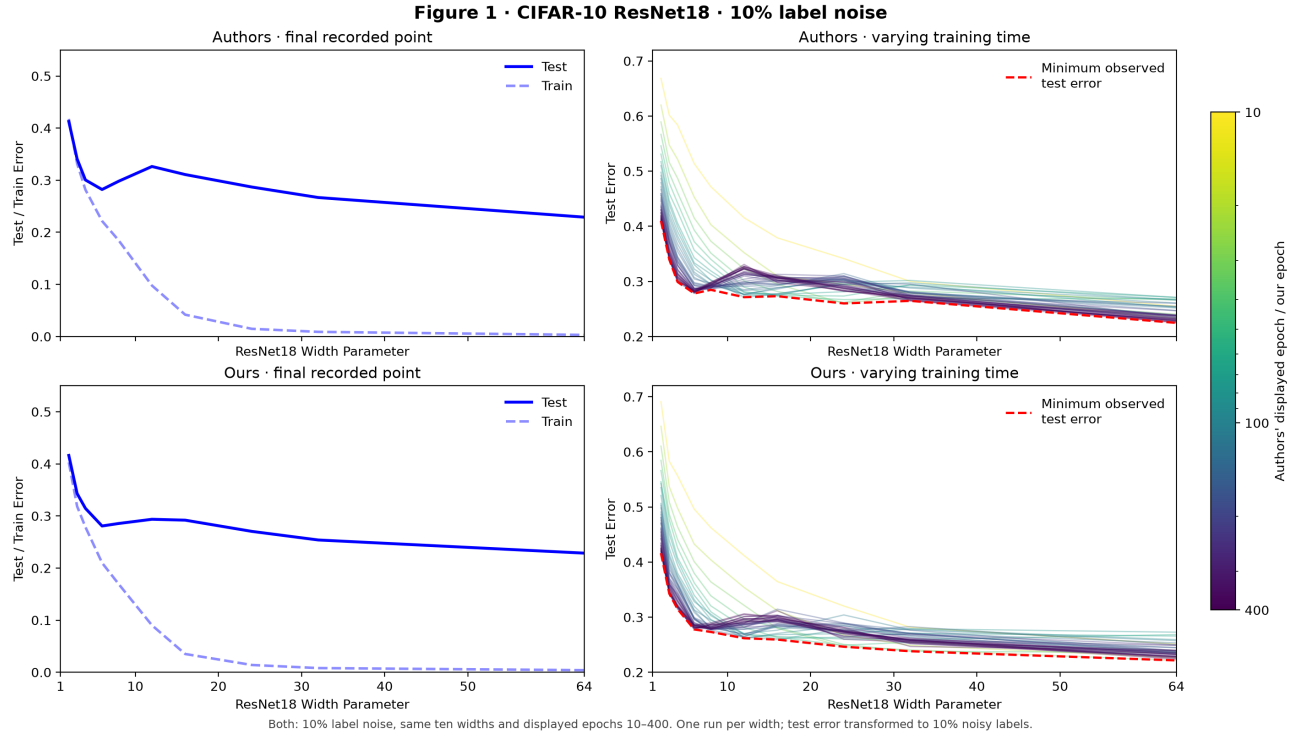
# 4. DISCUSSION AND LIMITATIONS

## 5. REPRODUCIBILITY AND CONTRIBUTIONS

The assignment code and result organization are available at the group’s GitHub repository. Detailed run coverage and deviations are recorded in the repository’s `EXPERIMENTS.md`. Group identification and individual contribution statements remain to be supplied by the group before submission.

| Dataset/model and setting             | Widths $k$                                                | Runs | Hours | Training horizon    |
|---------------------------------------|-----------------------------------------------------------|------|-------|---------------------|
| CIFAR-10 ResNet18, augmented          | 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                         | 30   | 4.53  | 400 Adam epochs     |
| CIFAR-100 ResNet18, augmented         | 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                         | 30   | 4.92  | 400 Adam epochs     |
| CIFAR-10 CNN, augmented               | 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 36, 40, 48, 64, 96, 128 | 48   | 1.45  | 50,000 SGD updates  |
| CIFAR-10 CNN, no augmentation         | 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                      | 33   | 0.74  | 50,000 SGD updates  |
| CIFAR-10 CNN, no augmentation         | 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                      | 33   | 2.13  | 400 Adam epochs     |
| CIFAR-100 CNN, clean, no augmentation | 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                      | 11   | 0.25  | 50,000 SGD updates  |
| CIFAR-10 CNN, augmented subsets       | 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64                      | 44   | 0.96  | 50,000 SGD updates  |
| Total                                 | One seed per cell                                         | 229  | 14.98 | Up to four runs/GPU |

**Table 1.** Completed sweeps, exact model widths, and measured UCloud job durations. Each width is run at label-noise probabilities 0, 10, and 20%, except the clean CIFAR-100 CNN sweep (0% only) and the subset sweep. For the subset sweep,  $p$  is the label-noise probability (10 or 20%) and  $n$  is the number of training examples (12,500 or 25,000): 11 widths  $\times$  2 noise rates  $\times$  2 subset sizes yield 44 runs. Other runs use all 50,000 training examples. Job IDs in row order are 12403634, 12403713, 12403687, 12403788, 12403936, 12403784, and 12403801. Durations are allocation wall times, rounded independently; overlapping model durations are not added to estimate GPU-hours.



**Fig. 1.** Noise-matched adaptation of the opening figure: authors’ released 10%-noise CIFAR-10 ResNet18 measurements above our 10%-noise results. Both panels use the same ten widths, 400 displayed epochs, and expected 10%-noise test-error transformation. Authors’ epochs follow their notebook convention of measurement index plus one; ours are recorded training epochs. Each side uses one released or measured run per width, not the five-trial mean of the original Figure 1. Red curves show the minimum observed test error over the displayed evaluations, not validation-based early stopping. The original opening figure instead uses 15% noise and 4,000 epochs [1].

## 6. REFERENCES

- [1] Preetum Nakkiran, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever, “Deep double descent: Where bigger models and more data hurt,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2021, no. 12, pp. 124003, 2021.

## A. COMPARISON OF EXPERIMENTAL SETTINGS

Table 2 compares the completed vision experiments one setting at a time. Page numbers refer to the 2021 journal article (32 pages), not an earlier arXiv version [1]. Released code paths refer to commit `0d653296`. The release contains model definitions, plotting notebooks, and a `README.md` linking result archives; it does not provide the vision training loop or data loader. Where neither the paper nor those files specify a detail, we identify it as unstated rather than infer a setting from the results.

**Table 2:** Our model, data, and training settings, their source locations, and differences from the stated protocol.

| Setting              | Our setup                                                                                | Paper setting and source location                                                                                                                                                                                          | Difference                                     |
|----------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Model</b>         |                                                                                          |                                                                                                                                                                                                                            |                                                |
| Vision architectures | Released ResNet18 and five-layer CNN; widths $[k, 2k, 4k, 8k]$                           | Same architectures: p. 17, Appendix B.1. Code: <code>models/resnet18k.py</code> and <code>models/mcnn.py</code> .                                                                                                          | Same definitions                               |
| Initialization       | Random, default PyTorch initialization                                                   | Default PyTorch initialization: p. 17, Appendix B.2, “Initialization”. Model files above instantiate PyTorch layers.                                                                                                       | Same rule                                      |
| Width grids          | Widths listed in Table 1                                                                 | Width sweeps: p. 17, Appendix B.1; exact arrays in linked archives, e.g. <code>cifar10-resnet18k-50k-adam/ks</code> and <code>pct-cifar10-mcnn-50000-p0-sgd-big/ks</code> . <code>README.md</code> supplies archive links. | Fewer widths                                   |
| <b>Data</b>          |                                                                                          |                                                                                                                                                                                                                            |                                                |
| Pixel preprocessing  | Divide by 255; no channel normalization                                                  | Not stated in Section 4 (p. 8) or Appendix B.2 (pp. 17–18). No released data-loader file.                                                                                                                                  | Not stated                                     |
| Label corruption     | Fixed Bernoulli mask; uniformly incorrect labels; fixed across augmentations             | Once-sampled, uniformly incorrect labels: p. 8, Section 4, “Label noise”. Fixed labels across augmentations: p. 17, Appendix B.2. No released implementation.                                                              | Same rule; seed not stated                     |
| Noise settings       | 0, 10, 20%; Figure 1 compares 10% on both sides                                          | Figure 4, p. 7, and experiment list in <code>README.md</code> ; original Figure 1, p. 4, uses 15%.                                                                                                                         | Same selected rates; original Figure 1 differs |
| Augmentation         | Padding 4, random crop 32, horizontal flip                                               | <code>RandomCrop(32, padding=4)</code> , <code>RandomHorizontalFlip</code> : p. 17, Appendix B.2, “Data-augmentation”. No released loader.                                                                                 | Same operations                                |
| Subsampling          | Seeded permutation; 12,500/25,000/50,000 training images                                 | Sample sizes: Figure 11(a), p. 13. Random subsampling: p. 19, Appendix B.4, “Sample-wise nonmonotonicity”. No released subsampling code.                                                                                   | Same sizes; permutation and seed not stated    |
| Test evaluation      | Clean test split; Figure 1 also transforms error to the expected noisy-test distribution | Clean test error except Figure 1: p. 8, Section 4. Transformation implemented in <code>intro_resnet_plot.ipynb</code> , “Simple Plot” and <code>plot</code> .                                                              | Same error convention                          |
| Train evaluation     | Fixed training labels, without augmentation                                              | Exact train-evaluation transform not stated in Appendix B.2 (pp. 17–18). No released evaluation loop.                                                                                                                      | Transform not stated                           |
| <b>Training</b>      |                                                                                          |                                                                                                                                                                                                                            |                                                |
| Loss                 | Cross-entropy                                                                            | Cross-entropy: p. 17, Appendix B.2, “Loss function”. No released training loop.                                                                                                                                            | Same loss                                      |

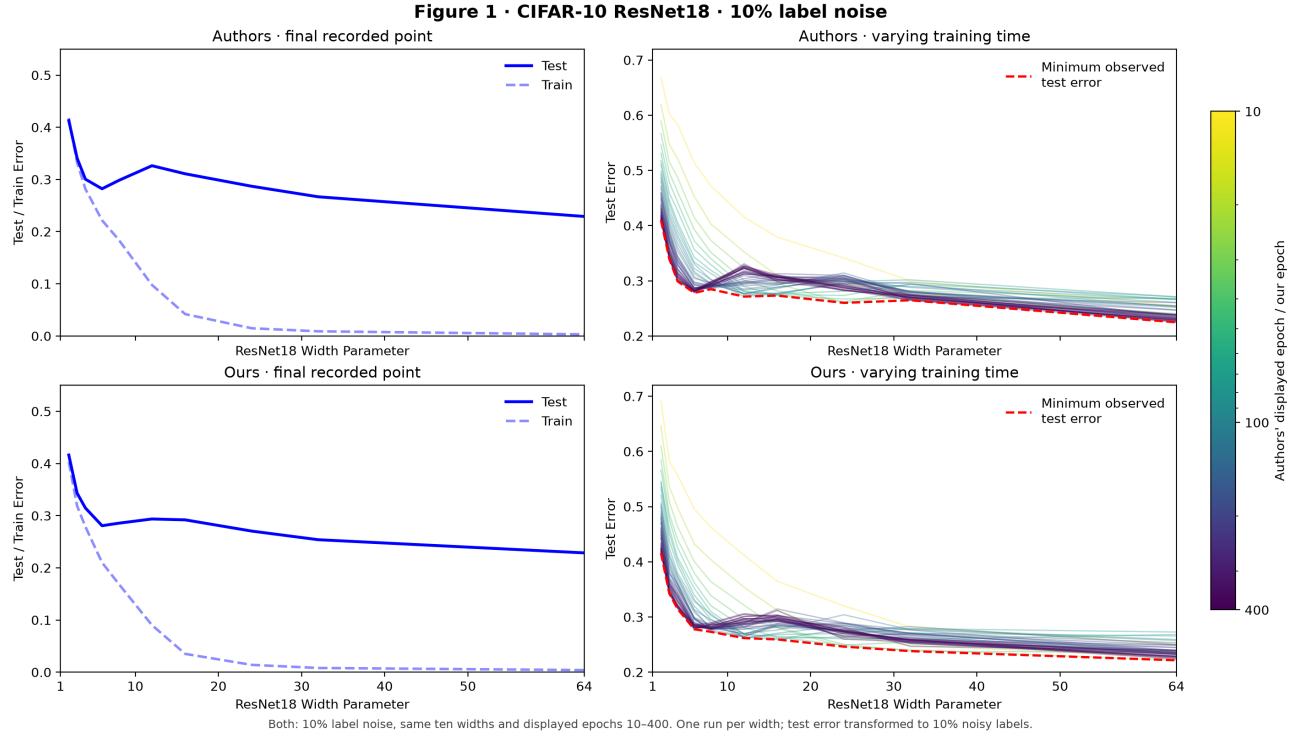
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Table 2 (continued)

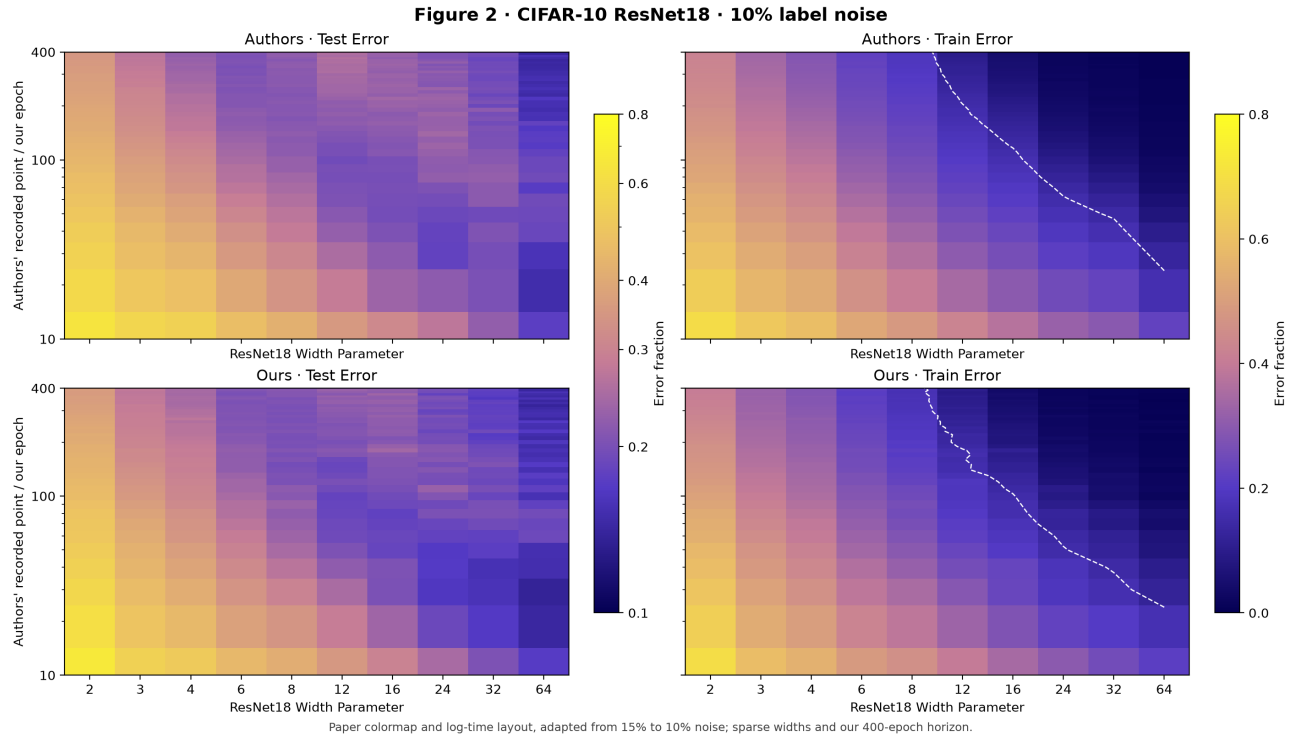
| Setting            | Our setup                                                 | Paper setting and source location                                                                                                                  | Difference                                  |
|--------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Batch size         | 128                                                       | 128: p. 18, Appendix B.2, “Batch size”. No released loader.                                                                                        | Same size                                   |
| Adam               | Constant LR $10^{-4}$ ; PyTorch defaults                  | Same settings: p. 17, Appendix B.2, “Optimization: Adam”. No released optimizer code.                                                              | Same settings                               |
| SGD                | LR 0.1, inverse-square-root; 512-step updates; momentum 0 | Same prescription: p. 18, Appendix B.2, “SGD” and “Learning rate schedule descriptions”. No released optimizer code.                               | Same prescription                           |
| Regularization     | No weight decay or dropout                                | None unless stated otherwise: p. 17, Appendix B.2, “Regularization”.                                                                               | Same setting                                |
| Training horizon   | Adam 400 epochs; SGD 50,000 updates                       | Adam 4,000 epochs; SGD 500,000 steps: p. 8, Section 4. CIFAR-100 CNN uses 1 million steps: Figure 20, p. 25, Appendix E.2.1.                       | 10% of stated horizon; 5% for CIFAR-100 CNN |
| Trials             | One seed (0) per condition                                | Five trials for Figures 1, 7, 11(a): p. 19, Appendix B.4. Figure 1 averaging is in <code>intro_resnet_plot.ipynb</code> .                          | One trial rather than five                  |
| Precision          | FP32                                                      | Arithmetic precision not stated in Section 4 or Appendix B.2. No released training loop.                                                           | Not stated                                  |
| Evaluation cadence | Adam: 10 epochs; SGD: 1,250 updates                       | <code>intro_ocean_dynamics.ipynb</code> labels ResNet indices plus one as epochs. CNN logging interval is not established by the paper or release. | No confirmed common cadence                 |
| Run infrastructure | Our loop and checkpoints; up to four runs per B200        | Checkpointing, concurrency and GPU scheduling not stated in Appendix B.2 or provided in the released files.                                        | Our implementation                          |

## **B. REPLICATION FIGURES: OURS COMPARED WITH THE AUTHORS**

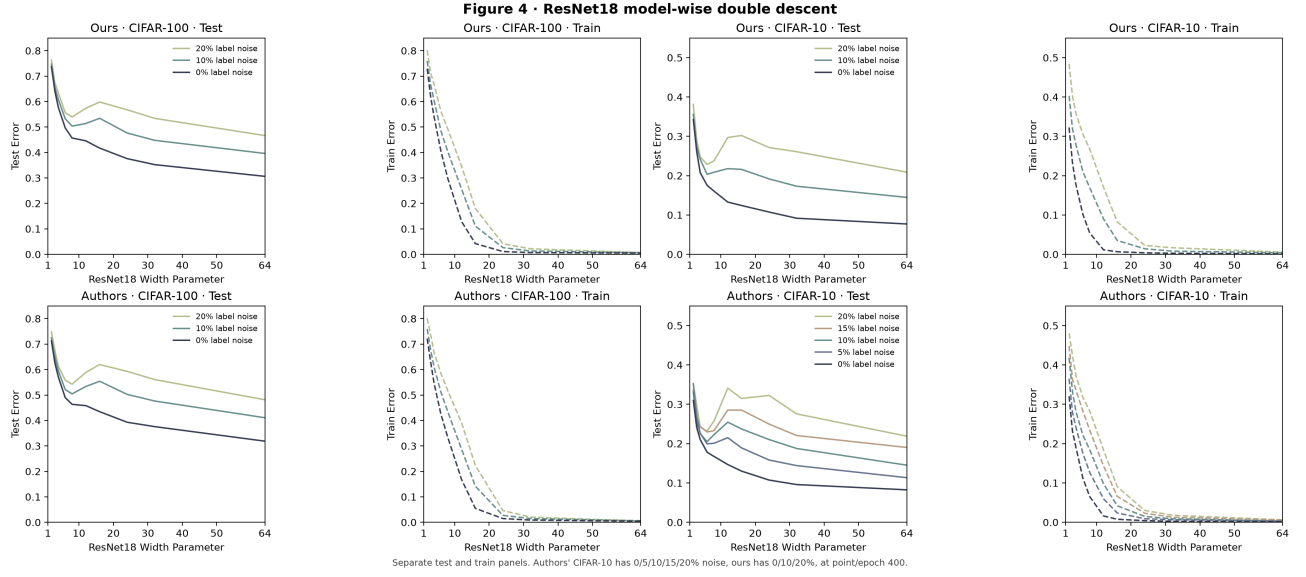
Every figure in this gallery places our replication results alongside measurements reconstructed from the authors' released data. There are no standalone galleries. Each caption identifies the shared settings and any differences in training horizon, trial count, or time coordinates. The Figure 1 comparison also appears in the main text. Released-data reconstructions are distinct from the original published artwork and are not additional experiments by our group [1].



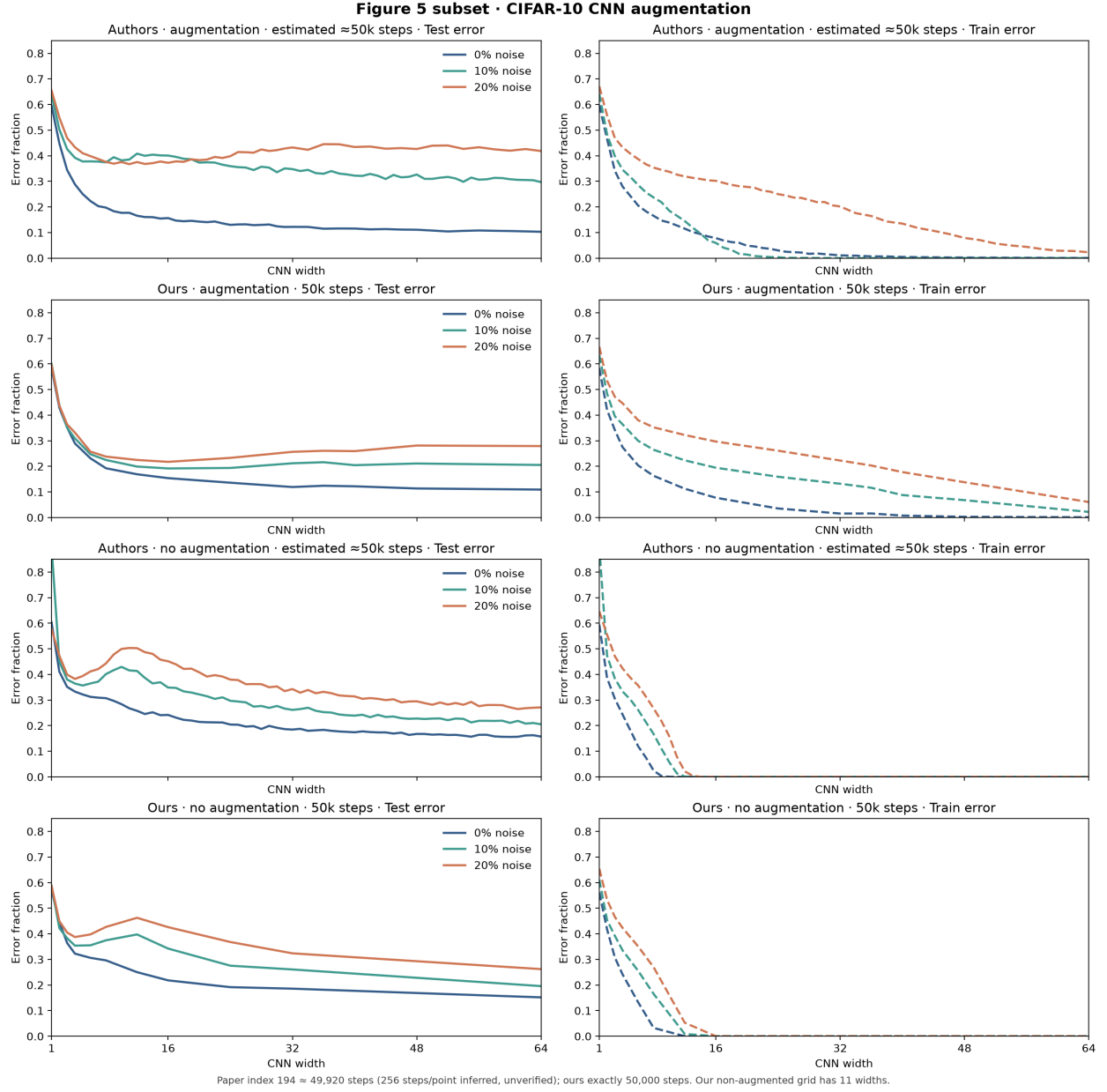
**Fig. 2.** Figure 1 adaptation: both authors and ours use 10%-noise CIFAR-10 ResNet18, the same ten widths and 400 displayed epochs, with one run per width. Authors' epochs follow the notebook convention; ours are recorded epochs. Test error uses the same expected noisy-test transformation.



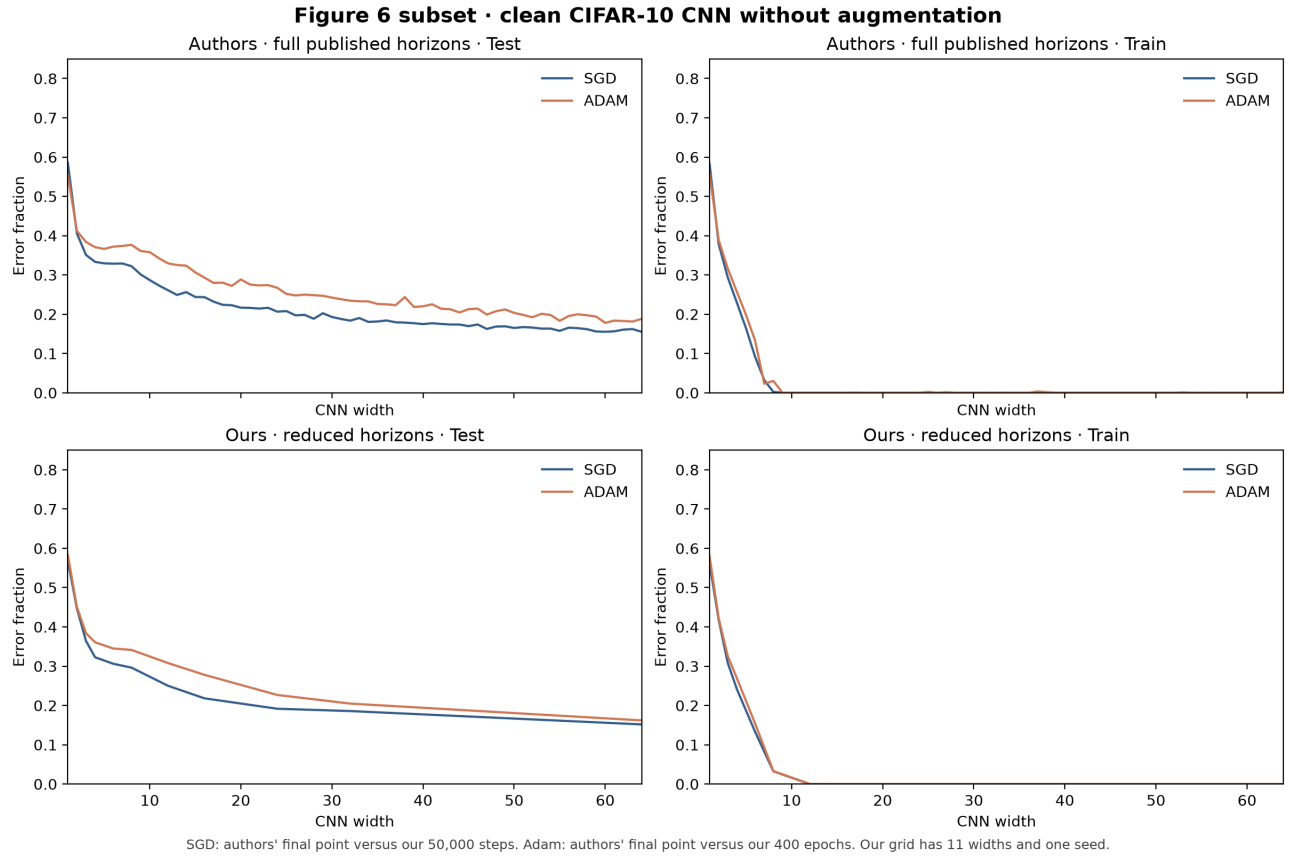
**Fig. 3.** Figure 2 adaptation: authors' and our CIFAR-10 ResNet18 train/test heatmaps at 10% noise, on the same ten widths and displayed horizon. Authors' epoch coordinates follow the released notebook convention; ours are recorded.



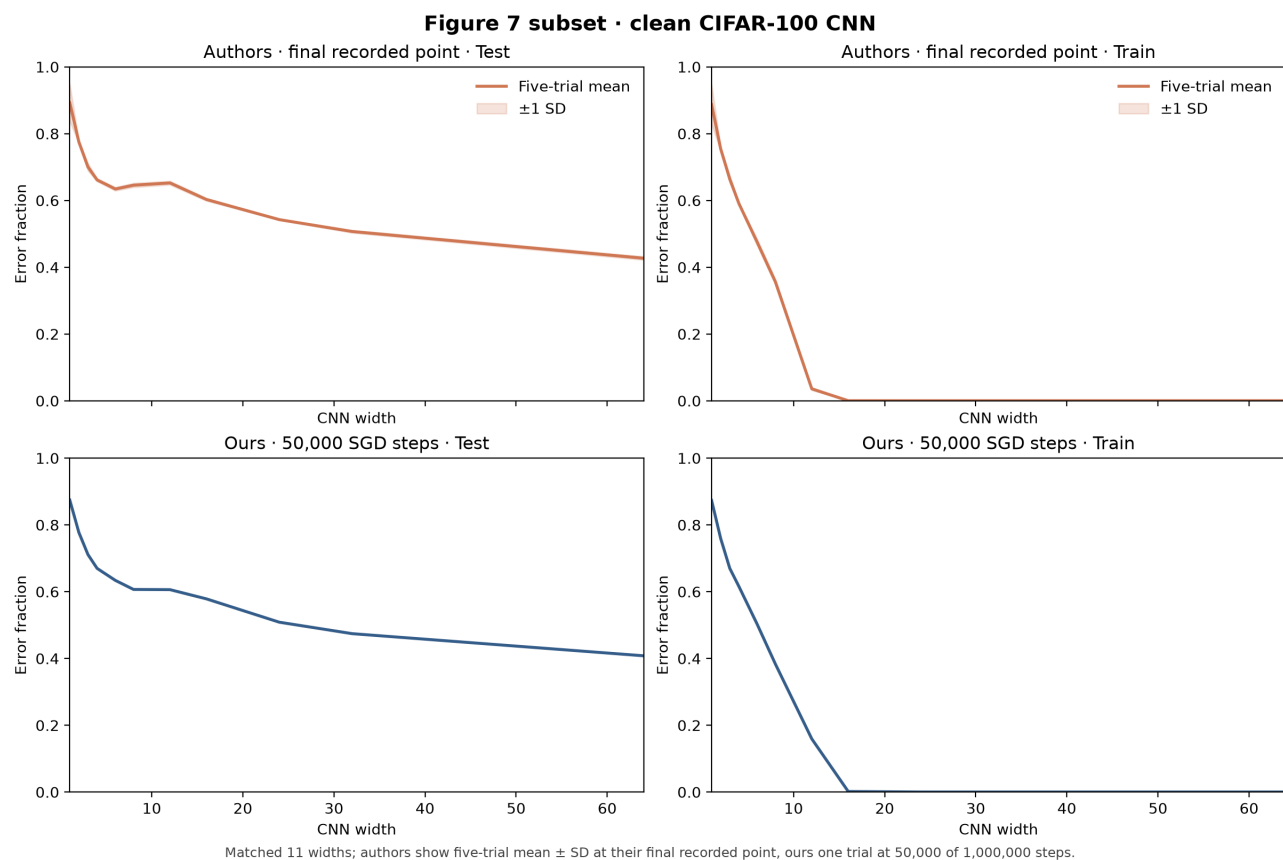
**Fig. 4.** Figure 4 adaptation: authors' and our CIFAR-10/CIFAR-100 ResNet18 error curves. Our 0/10/20%-noise results end at epoch 400; the released CIFAR-10 comparison also includes 5/15% noise.



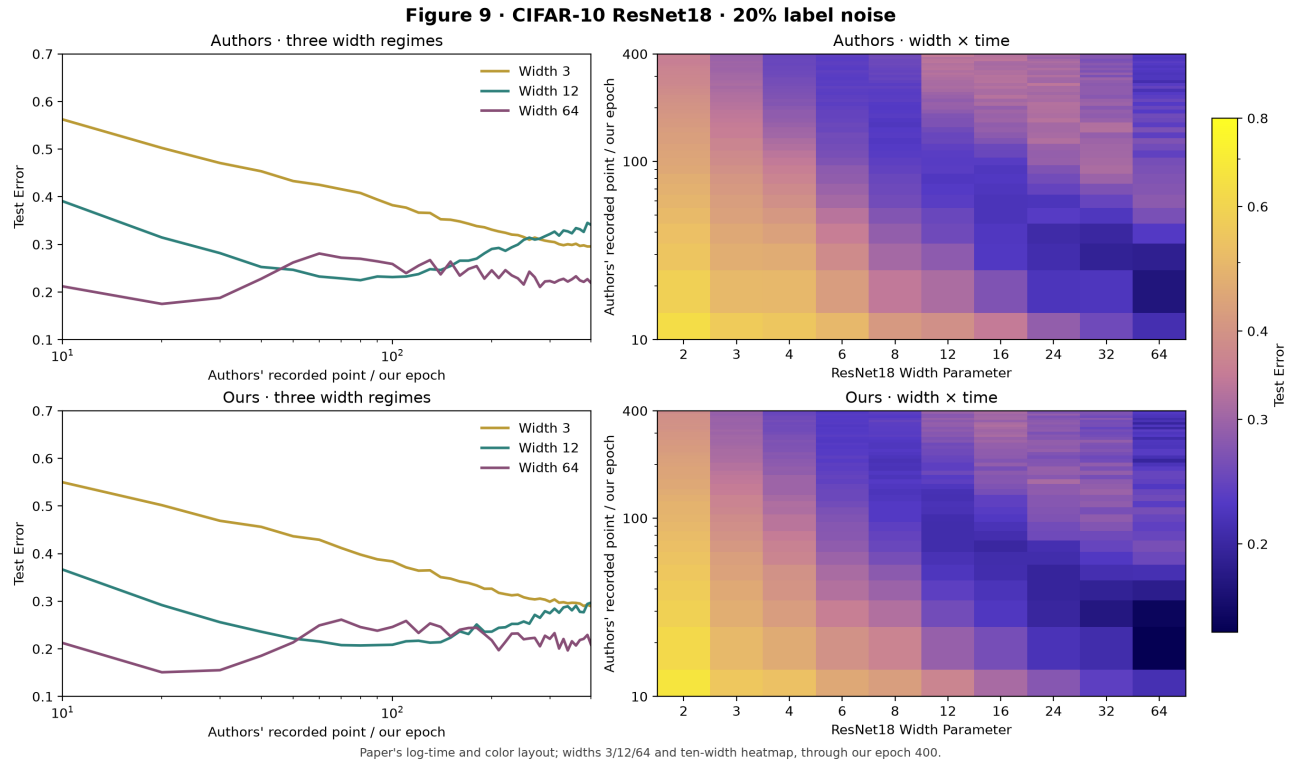
**Fig. 5.** Figure 5 subset: authors’ and our augmented and unaugmented CIFAR-10 CNN width curves. Ours end at 50,000 SGD updates; the authors’ nearby horizon uses an estimated, unverified step mapping.



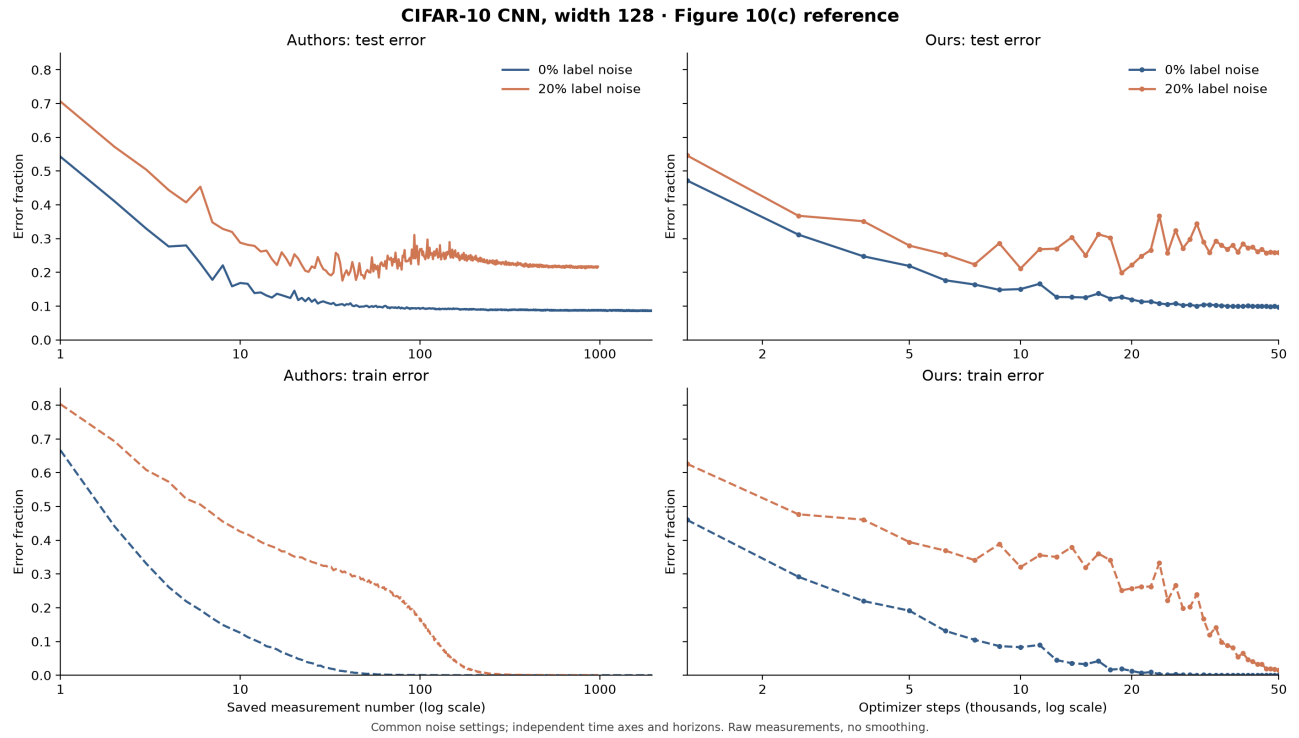
**Fig. 6.** Figure 6 subset: authors' and our clean, unaugmented CIFAR-10 CNN results with SGD and Adam. Authors' endpoints use full released histories; ours use 50,000 SGD updates or 400 Adam epochs.



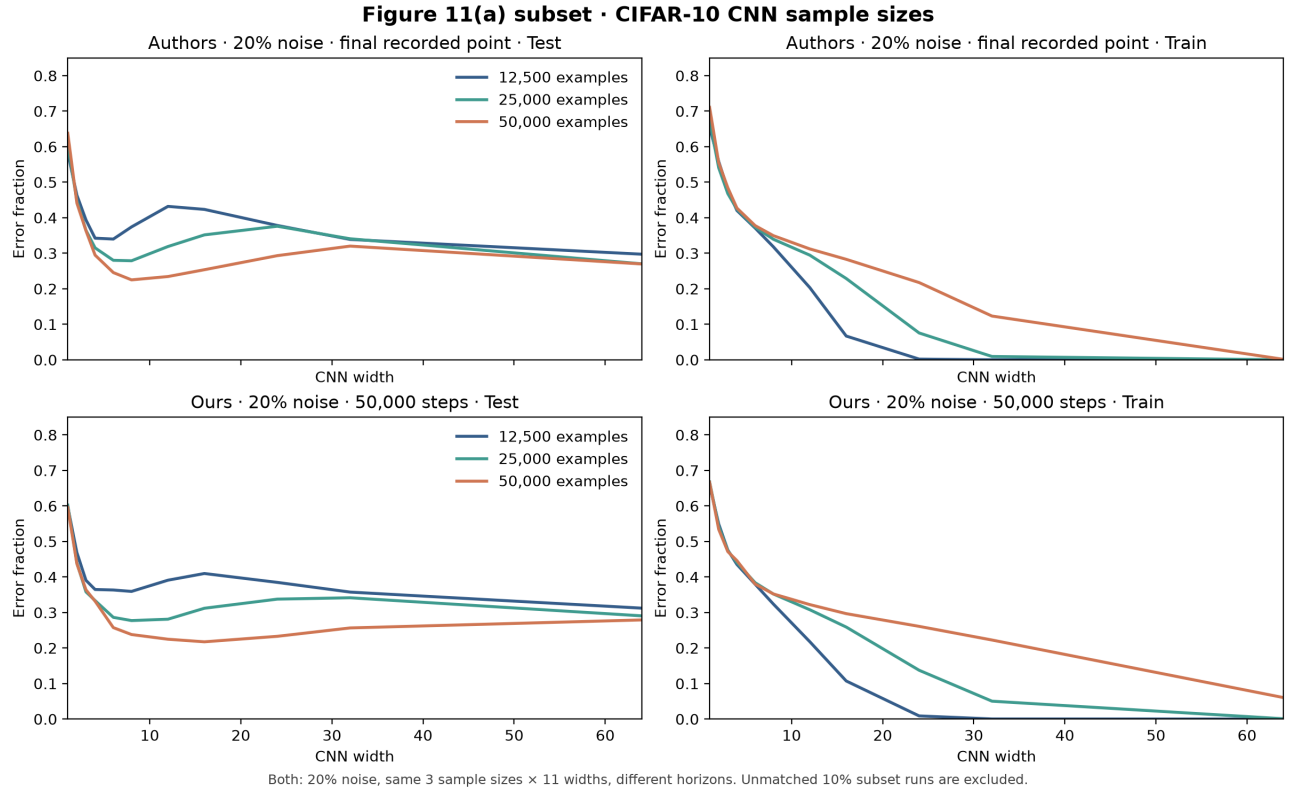
**Fig. 7.** Figure 7 subset: authors' and our clean CIFAR-100 CNN results at eleven common widths. The authors' five-trial mean and standard deviation use full histories; ours use one seed and 50,000 SGD updates.



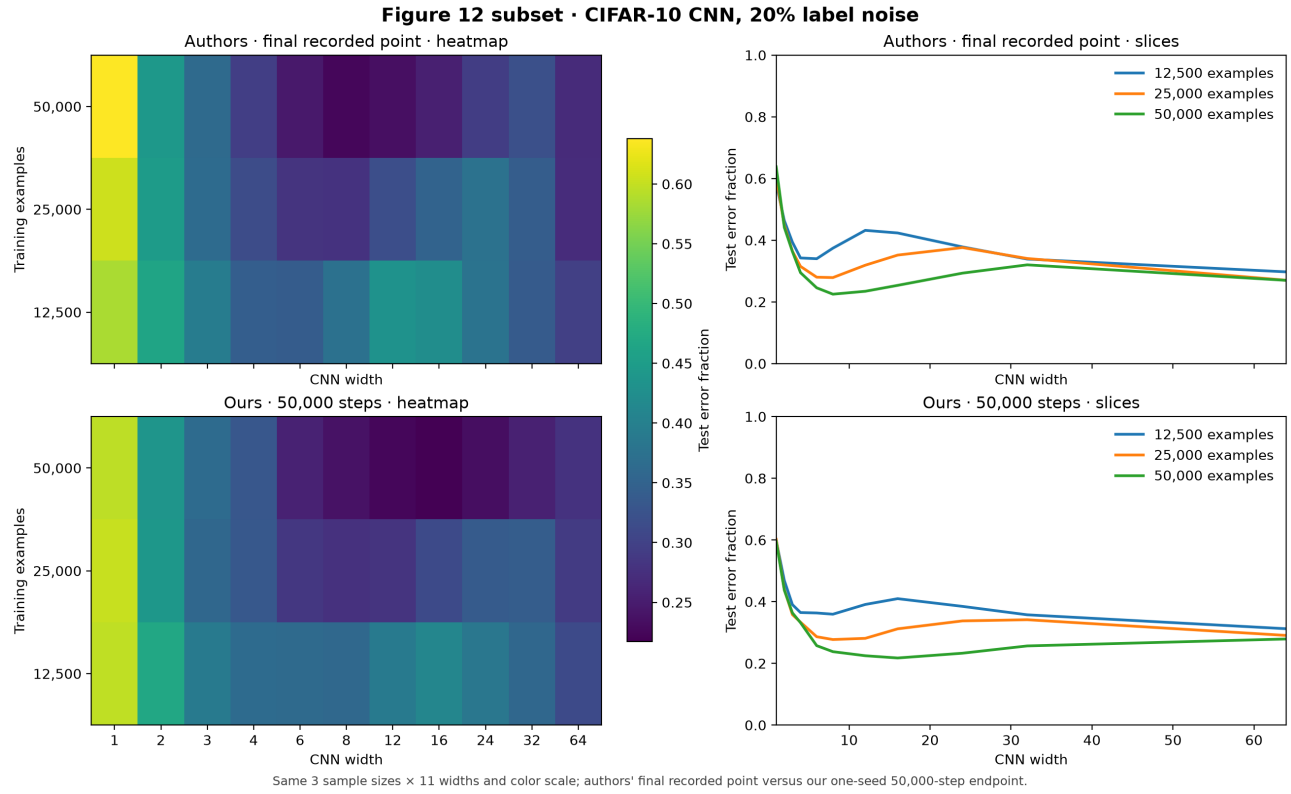
**Fig. 8.** Figure 9 adaptation: authors' and our 20%-noise CIFAR-10 ResNet18 trajectories and width-by-time heatmaps. Ours cover 400 epochs; the author reconstruction retains its labeled released coverage.



**Fig. 9.** Figure 10(c) subset: authors’ and our width-128 CIFAR-10 CNN trajectories at the common 0 and 20% noise settings. Authors’ clean full-data and 50,000-example/20%-noise references use saved measurement numbers; ours use recorded optimizer steps through 50,000 updates. Logging intervals and horizons are not matched. No smoothing is applied. The unresolved 10%-noise author source is excluded.



**Fig. 10.** Figure 11(a) subset: authors’ and our 20%-noise CIFAR-10 CNN sample-size curves, using 12,500/25,000/50,000 examples and eleven common widths. Authors’ endpoints use full recorded histories; ours stop at 50,000 updates. Our 10%-noise subset runs are excluded because no matching author source was identified.



**Fig. 11.** Figure 12 subset: authors' and our 20%-noise CIFAR-10 CNN sample-size heatmaps and slices. Both use three sample sizes, eleven common widths, and a common color scale; training horizons differ.